



Using dynamic software to explore linear relationships

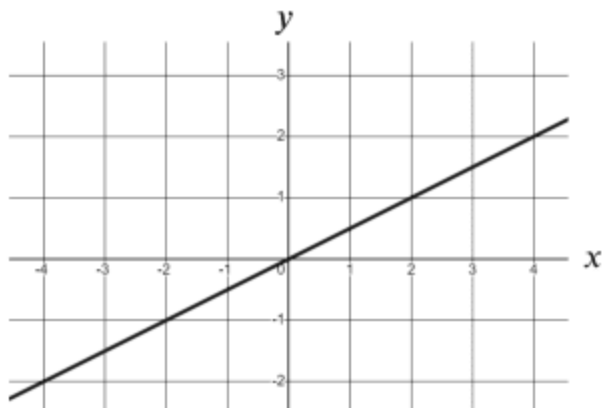
1 For relationships in the form $y = mx + c$ the value of m determines the _____. Tick **1** correct answer

- ☐ coordinates
- ☐ y-intercept
- ☐ x-intercept
- ☐ scale of the axes
- ☒ gradient

2 What of these statements will be true about the lines $y = 3x$, $y = 7x$, and $y = -2x$? Tick **2** correct answers

- ☒ They will all have a different gradient.
- ☐ They will be parallel lines.
- ☒ One and only one of the lines will have a negative gradient.
- ☐ They will all have a different y-intercept.

3 What is the equation of this line? Tick **1** correct answer



- ☐ $y = 2x$
- ☒ $y = \frac{1}{2}x$
- ☐ $y = x + 2$
- ☐ $y = x + \frac{1}{2}$

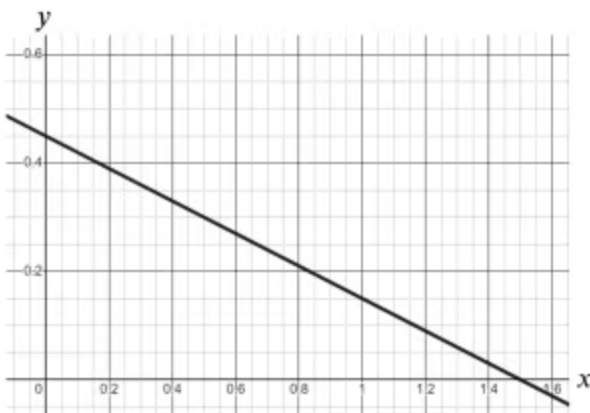


4 Which statements are true about the lines $y = x + 5$, $y = x + 9$, and $y = x + 0.5$? Tick 2 correct answers

- ☒ They have the same gradient.
- ☒ They are parallel lines.
- ☐ They have the same y-intercept.
- ☐ They have common coordinates.

5 The lines $y = 4 - 7x$ and $y = -0.1 - 7x$ have the same gradient. Fill in the blank

6 Calculate the equation of the line. Tick 1 correct answer



- ☒ $y = 0.45 - 0.3x$
- ☐ $y = 0.45 - 1.5x$
- ☐ $y = 0.5 - 0.3x$
- ☐ $y = 0.3 - 0.45x$

